

AFL Data Foundations — Data Dictionary & Target Specification Contract

Data Inventory: 43 AFL seasons (1983–2025), 20 clubs, 3,109 players, and 7,904 unique matches. **Data Quality Audit:** Stripped whitespace/tab prefixes; normalized team names ('W. Bulldogs' -> 'western bulldogs'); imputed corrupt negative disposals as K+HB; resolved sparse zero-encoding for goals/behinds; and purged duplicate biographical entries.

1. Relational Schema & Table Joining Contract

Table Name	Grain	Rows	Primary Keys	Join Path
team_matches_home_away_raw	Team-Match (2/game)	15,808	id / [date, team]	Pairs H to A on [date, year, round, clean(team)==clean(opp)]
afl_players_round_by_round_stats_raw	Player-Match (Box score)	274,089	id / [player_id, date]	Joins to matches on [date, year, round, team, opp]; to info on player_id
afl_players_seasonal_stats_raw	Player-Season-Phase	25,491	[player_id, year, team]	Joins to info on clean numeric player_id (regex stripped 'ID_')
afl_players_info_raw	Player Biographical	2,843	id	Master biographical dimension: name, debut, DOB, height, weight

2. Prediction Targets Specification (Day 2 Models Contract)

Target Name	Level	Type	Mathematical Definition / Formula	Justification / Downstream Role
home_margin (Primary)	Match	Regression	Score(Home) - Score(Away)	Retains victory magnitude; services spread/line betting; maps to prob via CDF.
home_team_win	Match	Binary	1 if Margin > 0, 0 if Margin < 0 (draws = 0.5)	Direct match-winner classification target (65 draws = 0.82% excluded/neutral).
top_disposal_getter	Player	Binary	1 if Disposals_i == max_match(Disposals)	Flags leading ball-winner across both squads in match; continuous = disposals.
top_goal_kicker	Player	Binary	1 if Goals_i == max_match(Goals) and Goals > 0	Flags leading goal-scorer in match; continuous = goals kicked.
fantasy_points	Player	Composite	3K + 2HB + 3M + 4T + 1HO + 6G + 1B + 1FF - 3FA	AFL Fantasy official scoring system (unbiased cross-position production metric).
player_impact_score	Player	Composite	Disp + 2*CP + 3*Clear + 4*I50 + 6G + 4T + 2*GA - 3*Clang	Brownlow/Champion Data composite measuring field-impact & territory gain.

3. Versioned Feature Dictionary (Strict Zero-Leakage Pipeline)

Feature Name	Window	Description	Predictive Signal & Findings
home/away_roll_win_5	5-game (shift 1)	Rolling 5-match win rate entering game	Form differential has r = 0.35 with margin; top quintile wins 80.2% vs bottom 40.6%.
roll_margin_5_diff	5-game (shift 1)	Net scoring margin differential (Home - Away)	Correlation r = 0.41 with final match margin; strong continuous signal.
home/away_win_streak	Expanding	Consecutive wins entering match (0 if loss/draw)	Measures team momentum and psychological winning form.
rest_diff	Pre-match lag	Days rest diff: home_days_rest - away_days_rest	+1 day rest advantage elevates home win rate from 58.1% to 60.6% (+1.8 pts margin).
is_interstate_match	Contextual	1 if visiting team traveled interstate, 0 for derby	Home win rate jumps from 55.8% (local derbies) to 63.1% against interstate travelers.
ladder_rank_diff	Round-entry	Pre-match ladder standing advantage (Away - Home)	Correlation r = 0.41 with margin after Round 4; 2.4 points margin per rank advantage.
player_roll5_disposals	5-game (shift 1)	Player rolling 5-match average disposals	Correlation r = 0.67 to actual match disposals; primary predictor for top disposals.
pos_archetype	Profile	Midfielder, Forward, Defender, Ruck, Utility	Midfielders avg 20.4 disp / 0.54 goals; Forwards avg 11.7 disp / 1.35 goals.

4. Reproducible Time-Based Split & Realistic Prediction Ceiling

Reproducible Split Protocol (src/features.py): get_time_based_split(df, split_year=2024, test_year=2025)

- **Train Set:** Seasons < 2024 (7,472 match rows, 254,216 player-game rows).
- **Validation Set:** Season 2024 (216 match rows, 9,937 player-game rows).
- **Hold-out Test Set:** Season 2025 (216 match rows, 9,936 player-game rows).

Temporal Leakage Rationale: In sports time-series, random cross-validation leaks future tactical adjustments, player breakout trajectories, and late-season momentum into earlier-round predictions. Real-world evaluation must strictly mirror production: training on historical seasons and evaluating forward in time.

Realistic Prediction Ceiling: AFL is an inherently high-variance contact sport played with an elliptical ball across varying oval dimensions, vulnerable to in-game injuries, microclimatic weather shifts, and umpire adjudications. Top-tier industry models (Squiggle consensus, Matter of Stats, Footy Forecaster) achieve a realistic ceiling of **67% to 72% match accuracy** (log loss ~0.58–0.62). Any model reporting >80% or 'near-perfect' accuracy is an undeniable red flag indicating target leakage (e.g., in-match count stats like inside-50s) or forward lookahead bias.

Artifact Version: afl_match_features_v1.parquet (0.66 MB) · afl_player_match_features_v1.parquet (5.44 MB) · Git Branch: main